

Ecological Models

CONSIDERATIONS FOR TEST DESIGN TO ACCOMMODATE ENERGY-BUDGET MODELS IN ECOTOXICOLOGY: A CASE STUDY FOR ACETONE IN THE POND SNAIL *LYMNAEA STAGNALIS*

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Abstract: Toxicokinetic–toxicodynamic (TKTD) modeling offers many advantages in the analysis of ecotoxicity test data. Calibration of TKTD models, however, places different demands on test design compared with classical concentration–response approaches. In the present study, useful complementary information is provided regarding test design for TKTD modeling. A case study is presented for the pond snail *Lymnaea stagnalis* exposed to the narcotic compound acetone, in which the data on all endpoints were analyzed together using a relatively simple TKTD model called DEBkiss. Furthermore, the influence of the data used for calibration on accuracy and precision of model parameters is discussed. The DEBkiss model described toxic effects on survival, growth, and reproduction over time well, within a single integrated analysis. Regarding the parameter estimates (e.g., no-effect concentration), precision rather than accuracy was affected depending on which data set was used for model calibration. In addition, the present study shows that the intrinsic sensitivity of snails to acetone stays the same across different life stages, including the embryonic stage. In fact, the data on egg development allowed for selection of a unique metabolic mode of action for the toxicant. Practical and theoretical considerations for test design to accommodate TKTD modeling are discussed in the hope that this information will aid other researchers to make the best possible use of their test animals. *Environ Toxicol Chem* 2014;33:1466–1475. © 2014 SETAC

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INTRODUCTION

Toxicity tests, like any experimental test, are defined by their test design. Differences between test designs for a given species and toxicant may lead to large differences in the test results. Reviewing toxicity tests for the gastropod *Lymnaea stagnalis* exposed to tributyltin, Giusti et al. [1] showed that different protocols led to different values for the no-observed-effect concentration (NOEC) and the estimated concentration causing $x\%$ effect (EC x), and even to conflicting conclusions about the adverse effects on reproduction for snails exposed to field-relevant concentrations. Reflecting the importance of test design, standard toxicity test protocols have been produced for a suite of test species; they define tailored test designs that ensure the reproducibility of toxicity tests within a species, which is essential when their results are to be used in ecological risk assessment.

A test design combines fixed experimental conditions and fixed experimental effort. Experimental conditions comprise environmental factors that should be controlled during the test to ensure the well-being of the animals in the absence of toxic stress (e.g., feeding conditions, light and temperature, water quality, density). These factors are thus mainly species dependent. Experimental effort comprises the number and range of test concentrations, the number of replicates, the number of animals per replicate, the test duration, the endpoints to be measured, and the frequency of these measurements over the test duration. The experimental effort is a complex component, as the optimal test

design depends on the species, the toxicant, and especially on the foreseen analysis of the resulting data.

Ecotoxicity test data are routinely analyzed using descriptive approaches to derive a summary statistic such as NOEC or EC x . Current standard test protocols have been mainly designed to satisfy data analysis by hypothesis testing and concentration–response modeling. However, other modeling approaches are possible and are recognized as valid in standard procedures for data analysis [2,3], such as mechanistic models that describe the toxicokinetics and toxicodynamics of chemicals (i.e., toxicokinetic–toxicodynamic [TKTD] models). These models offer many advantages for the analysis of ecotoxicity tests [4,5]. Most importantly, they make optimal use of the information from the test data (i.e., integrating observations over time for different endpoints), and allow extrapolation to other environmental conditions and other exposure patterns. Furthermore, they hold promise for reducing the number of test animals and the costs of toxicity tests, and for generating useful information to define risk mitigation measures. This makes TKTD models very appealing to risk assessors and stakeholders [6,7]. Such models place different demands on test design, but guidance on the most appropriate design to calibrate this kind of model is not yet available. This lack of guidance could be one of the major reasons why these models are not widely applied yet in ecotoxicology or environmental risk assessment.

The present study aims at providing useful complementary information for the analysis of toxicity tests data using a TKTD model. In this context, we exposed the pond snail *L. stagnalis* to a narcotic toxicant (acetone). The experimental test was designed to optimize the calibration of a simple TKTD model from a small number of test animals, thus reducing experimental costs. We chose to work on the pond snail because standard protocols with this species are currently under development [8]

All Supplemental Data may be found in the online version of this article.

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under the Organization for Economic Cooperation and Development (OECD) test guidelines development program. A narcotic chemical was selected because of its relatively simple mechanism of action and to obtain results that would be relevant for a large number of toxicants. Among the existing TKTD models, only models that are based on dynamic energy budget (DEB) theory [9] have been included in guidances for the analysis of ecotoxicity data published by the International Organization for Standardization [2] and the OECD [3] (although so far they have rarely been applied in risk assessment).

To promote the use of effect models, both for scientific and for risk assessment purposes, it is essential that models are not too complex, so that ideally it would be possible to understand and use them without a background in modeling. Therefore, we used a simplified DEB model called DEBkiss [10] for the analysis of toxicity test data. Through a stepwise procedure, we evaluated the benefits of using different types of data in the model calibration. We explored how standard toxicity test data (i.e., survival, size, and fecundity of adults) and additional data (i.e., the same endpoints in juveniles, and exposure–recovery patterns in both juveniles and adults) affect model calibration. The optimal data set for calibration of the DEBkiss model is discussed in relation to the data already available from standard tests to scrutinize the costs and benefits of additional experimental effort. Finally, we used the calibrated model to predict independent data for embryonic development (hatching success, hatching time, and snail size at hatching under toxic stress) to test the model's ability to describe this life-cycle stage as well.

MATERIALS AND METHODS

Conceptual description of the DEBkiss model

A range of practical models has been derived from DEB theory, but most of the ecotoxicological applications rely on a simplification often referred to as DEBtox [11–13]. Because it contains fewer details about the animal's physiology than the standard DEB animal model [14], it is easier to parameterize and its calibration is less data demanding, which better suits the

available data from toxicity tests. In the present study we apply a more recent simplification, called DEBkiss [10]. In contrast to DEBtox, this model sports an explicit mass balance, which allows, for example, the implementation of starvation rules that respect mass and energy conservation. Furthermore, DEBkiss does not divide biomass into a reserve and structure component, and explicitly includes the embryonic stage. This model was earlier shown to provide a good description of the life cycle of *L. stagnalis* [10]. The structure, assumptions, and equations of the DEBkiss model used in the present study are presented in the Supplemental Data.

In TKTD frameworks, a toxicant first must enter the organism and reach a target site (toxicokinetics), where it will then exert toxic effects (toxicodynamics). Here we use a scaled first-order, one compartment toxicokinetic model accounting for the effects of changes in body size and losses due to reproduction (Supplemental Data). For embryos, toxicokinetics is likely different than for juveniles and adults, because of the different shape and nature of the interface between organism and exposure medium (the egg mass and egg membrane). Here, we simply assume instantaneous steady state between the egg and the environment, which is defensible in this case given the small size of the eggs and the fact that they were removed from the egg mass, and because of the rather rapid kinetics of acetone (a relatively high value of the elimination rate; see Table 1).

For toxicodynamics, we assume that the scaled internal concentration affects one or more of the DEBkiss model parameters that describe metabolic processes (e.g., assimilation, somatic maintenance, and maturation). An effect on a particular process directly translates into a specific change in the physiological responses of the snail over time (e.g., growth and reproduction). The affected metabolic process(es), referred to as the metabolic mode of action (MMOA, [15]) can thus be selected on the basis of the observed patterns in life-history traits. For a particular MMOA, a linear-with-threshold relationship is assumed between the scaled internal concentration and the affected metabolic process; the threshold is referred as the no-effect concentration (NEC). If several metabolic processes are

Table 1. Parameter values as estimated based on best fit to test data with *Lymnea stagnalis*^a

	Symbol	Value	Unit
Metabolic model parameters			
Specific maximum assimilation rate	J_{Am}^a	0.187 (CI: 0.170–0.208)	mg mm ⁻² d ⁻¹
Specific maintenance rate	J_M^a	0.0106 (CI: 0.00956–0.0119)	mg mm ⁻³ d ⁻¹
Allocation fraction to soma	κ	0.619 (CI: 0.606–0.627)	–
Shell length at maturation	L_{wp}	25.2 (CI: 25.0–25.6)	mm
Blank hazard rate	h_0	1.11×10^{-3} (CI: 0.184×10^{-3} – 3.39×10^{-3})	d ⁻¹
Shape correction coefficient	δ_M	0.367 (SD: 0.0293)	–
Dry wt density of structure	d_V	0.105 (SD: 5.90×10^{-3})	mg mm ⁻³
Dry wt of a freshly laid egg	W_{B0}	0.15 ^b	mg
Yield of egg buffer on assimilates	y_{BA}	0.95 ^b	mg mg ⁻¹
Yield of structure on assimilates	y_{VA}	0.8 ^b	mg mg ⁻¹
Scaled functional response	f	1 ^b	–
Toxicological model parameters			
Elimination rate constant	k_e	0.928 (CI: 0.754–1.16)	d ⁻¹
NEC for survival	c_{0s}	13.5 (CI: 12.6–13.8)	mL L ⁻¹
Killing rate	b	0.0349 (CI: 0.0193–0.0574)	L mL ⁻¹ d ⁻¹
NEC for metabolic effects	c_0	0.879 (CI: 0.807–0.905)	mL L ⁻¹
Tolerance concentration	c_T	26.2 (CI: 25.2–26.8)	mL L ⁻¹
Yield of assimilates on structure	y_{AV}	0.397 (CI: 0.299–0.533)	mg mg ⁻¹

^a Parameters δ_M and d_V were directly calculated from length data of both juvenile and adult snails. The values of the toxicological parameters are calculated using nominal concentrations for exposure.

^b Parameter values are not calibrated but fixed to the suggested values [10].

CI = 95% confidence interval; SD = standard deviation; NEC = no-effect concentration.

altered by a toxicant, then each of them can have its own value for the no-effect concentration. However, we depart from the assumption that the observed changes in life history result from an effect on a single process. The internal concentration also increases the probability for the individual to die. This is modeled through a hazard rate, which is also linked to the scaled internal concentration using a linear-with-threshold relationship (fully equivalent to the case of simple stochastic death in the general unified threshold model for survival [16]). The NEC for effects on mortality can be different from the one for metabolic effects (which affects sublethal endpoints).

Test species and experimental conditions

Snails (*L. stagnalis*, RENILYS strain) were cultured at the French National Institute for Agricultural Research Experimental Unit of Aquatic Ecology and Ecotoxicology (Rennes, France), under conditions previously described [17]. The culture medium was dechlorinated, charcoal-filtered tap water with the following physicochemical characteristics: pH 7.4 ± 0.2 , conductivity $428 \mu\text{S cm}^{-1} \pm 4 \mu\text{S cm}^{-1}$, and dissolved oxygen $6 \text{ mg L}^{-1} \pm 1.2 \text{ mg L}^{-1}$. Snails were reared at a temperature of $20.9^\circ\text{C} \pm 1.4^\circ\text{C}$ and a photoperiod of 14-h:10-h light:dark. Juvenile (mean shell length of $18.6 \text{ mm} \pm 0.5 \text{ mm}$; no observed reproduction in the culture) and adult snails ($29.3 \text{ mm} \pm 1.4 \text{ mm}$) were sampled from the laboratory culture and placed individually into glass beakers filled with 200 mL of the test medium (i.e., culture medium spiked with acetone). Snails remained in the test medium for the whole time during exposure, so that snail behavior did not affect the hypothesis of constant exposure. The test medium was renewed twice a week to maintain its physicochemical properties and to avoid fouling of test vessels (which were additionally cleaned with soft paper tissue at each test medium renewal). Animals were fed ad libitum with rinsed fresh organic lettuce. Prior to exposure, all snails were acclimated for 5 d in test beakers, which ensured that they had recovered from sampling stress.

Test substance

Acetone CHROMASOLV (CAS number 67-64-1, purity $\geq 99.9\%$; Sigma-Aldrich) was used as the test substance. Because this chemical is widely used in toxicity tests as a carrier solvent for sparingly soluble substances, gaining greater insight into its unintended effects on organisms is desirable. Acetone was diluted in the culture medium by pouring the appropriate volume of acetone into each test beaker, using a micropipette, after removal of an equivalent volume of culture medium. The following nominal concentrations of acetone were used: 0.1 mL L^{-1} , 0.3 mL L^{-1} , 0.9 mL L^{-1} , 3 mL L^{-1} , 9 mL L^{-1} , 14 mL L^{-1} , and 18 mL L^{-1} . They were chosen based on the results of Bluzat et al. [18] and of pilot experiments we performed, aiming for a range from small to large effects on survival and reproduction. Acetone is a highly volatile chemical, and therefore, deviations from nominal concentrations were expected to occur between 2 renewals of the test medium. We expect volatilization to be a first-order process, which means that the relative differences between the treatments are preserved. For simplicity, we will always refer to the nominal concentrations in the present study.

Tests designs and endpoints

To obtain calibration data for the DEBKiss model, we conducted 4 partial life-cycle toxicity tests. The tests were performed simultaneously on groups of juvenile and adult snails. Tests with juveniles started with snails that were close to their size at maturity ($\sim 20 \text{ mm}$). They became adults (onset of egg

production) during the tests, but we continue to refer to them as juveniles in the present study. In addition, an embryo toxicity test was performed.

Exposure tests. Five juvenile and 5 adult snails were exposed in isolation to acetone concentrations of 0.1 mL L^{-1} to 18 mL L^{-1} for 56 d. For the control group, 25 isolated snails were used instead of 5 to obtain more precise estimates for the metabolic parameters of DEBKiss. Survival was monitored daily. Dead snails were removed from the beakers and stored at -20°C for dry-weight measurement. Growth was assessed from shell length, measured once a week using a digital caliper, and from dry weight of soft bodies, determined only once at the end of the test (or when an individual died). Reproductive investment per snail was quantified through the cumulative dry weight of egg clutches. Test beakers were checked daily for the presence of egg clutches. They were carefully removed from the glass surface with a sharp-edged spoon and placed individually into plastic microtubes. The samples were stored in a freezer at -20°C prior to being freeze-dried for 24 h and weighed. To decrease the potential impact of isolation on the snails' reproductive performance, we allowed them to mate regularly during the tests. Snails were coupled (1 juvenile and 1 adult snail, always using the same couple) twice a week for 8 h, which duration sufficed to allow snails to mate [19]. Data sets obtained from juvenile and adult exposure tests are hereafter referred as Je and Ae, respectively.

Recovery tests. Four adult and 4 juvenile snails were exposed individually to acetone concentrations of 9 mL L^{-1} to 18 mL L^{-1} for 14 d, followed by a 42-d recovery period in clean medium. The control snails from the exposure test also served as controls for the recovery test. The same endpoints were followed as for the exposure test. Data sets obtained from juvenile and adult recovery tests are hereafter referred as Jr and Ar, respectively.

Embryo test. To obtain data on embryonic development, we exposed snail eggs to a range of acetone concentrations from 0.02 mL L^{-1} to 18 mL L^{-1} according to the method developed by Bandow et al. [20]. Clutches, less than 24 h old, were collected from the laboratory cultures from adults of homogeneous age. Eggs from all collected clutches were extracted from their jelly-masses using a Pasteur pipette and placed into a Petri dish. Next, 12 eggs were randomly selected per treatment and placed into a 12-well plastic plate containing 10 mL of acetone exposure medium per well. The test duration was 21 d, the test medium was renewed twice a week, the photoperiod was 14-h:10-h light:dark, and the temperature was $22.4^\circ\text{C} \pm 1.1^\circ\text{C}$. The following endpoints were monitored: time to hatch (checked daily after the first wk of exposure), shell length at hatching, and hatching success (embryos that did not hatch after 21 d were considered dead).

Stepwise model calibration strategy

Shell length is used as quantifier for body size of the snails over time, as it can be determined nondestructively. However, the state variable for body size in the DEBKiss model is dry body mass. Shell length is translated into dry body mass using a shape correction coefficient (δ_M) and the dry density of body tissue (d_V ; see Supplemental Data). The shape correction coefficient was determined as a mean value from 50 individuals (juveniles and adults) of the control group. Dry-weight density of structure was obtained from an independent experiment performed in our laboratory and was calculated as the mean value of 60 observations from nonexposed snails (Table 1).

The metabolic parameters of DEBKiss (Table 1) were estimated using the data from all control snails (juveniles and

adults) simultaneously. Indeed, experiments on adults alone do not provide sufficient information to identify the metabolic model parameters, owing to a lack of information on the growth rate and the start of reproduction. The only parameter that differed between the 2 age classes was the initial shell length. The initial size was taken as the mean of all shell sizes in each age class, and fixed in the analysis. The 2 yield coefficients (y_{BA} and y_{VA}) cannot be independently estimated from these data and were fixed to recommended values [10]. The complete set of estimated metabolic model parameters was subsequently fixed in the analysis of the data for the exposed snails.

Toxicological parameters were estimated by simultaneously fitting the data on survival, growth (shell length and dry soft body mass), and reproduction for all exposed snails (using nominal concentrations for exposure). To account for the variability of snail sizes between treatments, initial shell size was fixed to the mean size for each treatment, as described for control snails. The observations on dry soft body mass of snails indicated that animals shrank during exposures to higher concentrations of acetone. This implies that body tissue is used to pay maintenance costs (see Supplemental Data); therefore, we were able to estimate the yield of assimilates on structure, y_{AV} . To avoid unrealistic estimates, we restricted y_{AV} to between 0.05 and 0.95.

The model was implemented in Matlab 2010b, and parameter values were estimated by maximizing the overall likelihood function [11] (using square-root transformation of data for the continuous endpoints). The 95% confidence intervals on parameter estimates were generated by profiling the likelihood function [11].

To investigate what experimental design would be most suitable for model calibration, we calibrated the model using the data either from each experiment separately or in combination. This generated 9 calibration and model parameter sets: Je, Jr, JeJr, Ae, Ar, AeAr, JeAe, JrAr, and JeJrAeAr. To test how the different parameters sets predict different data sets, we used each parameter set to predict the observations in each data set (comparing the log-likelihood value of the prediction with the best possible value for that data set). The most likely MMOA of acetone was selected based on the best model fit to all data simultaneously (JeJrAeAr), and fixed for the 8 subsequent calibration sets. Fitting the model to different calibration sets suggested different possible MMOAs. However, the most probable MMOA is the one that allows integrating effects on all monitored endpoints, as reflected by the JeJrAeAr calibration set. The correctness of the MMOA selected using this calibration set was confirmed by the observed effects on embryo development.

The data for effects of acetone on embryonic development were not used to fit the model, but only to test whether the parameter values from a fit on juvenile and adult data suffice to predict toxicity to the developing egg. In light of previous model analyses for pond snail eggs [10], we assumed an initial lag phase of 2.5 d in embryonic development and tuned the specific assimilation rate of the embryo to match the observed hatching time in the control. The initial dry mass of a single egg (W_{B0}) was taken as 0.15 mg [10].

RESULTS AND DISCUSSION

Fitting control data

The DEBkiss model provided an excellent fit to all of the data for all control snails simultaneously (Figure 1). This supports the assumption that juveniles and adults can be described by the same parameter set for basic physiology (Table 1). The tight

confidence intervals show that the parameter values are well defined from the available data.

The DEBkiss model was previously calibrated for the pond snail by Jager et al. [10] using other experimental data from the same laboratory as for the present study. In contrast to the earlier findings, snails in the present study start to invest in reproduction at a higher body dry mass (79 mg vs 67 mg, which corresponds to 25.2 mm and 21.4 mm shell length, respectively). This discrepancy might relate to differences in experimental setup between the studies. Indeed, snails were maintained in groups of 5 previously [10], whereas they were kept isolated in the present study. *Lymnaea stagnalis* is a simultaneous hermaphrodite that reproduces preferably by outcrossing. However, in the absence of conspecifics, selfing occurs. Kept in isolation, snails may delay the egg laying [21], which might be a strategy to ensure lower inbreeding depression [22]. Snails in the present study were maintained in isolation but were allowed to mate twice a week, which might constitute a suboptimal mating frequency. Differences in soft body dry weight and size at maturation might also result from variation in life-cycle traits among snails (Supplemental Data, Figures S2–S4).

Fitting all treatments together

To select an MMOA and to obtain a reference set of toxicity parameters, we fitted the model to data from all treatments together, fixing the metabolic parameters to the values obtained in control conditions (Table 1). The corresponding model fit for the exposure tests only is shown in Figure 1A to H; fits on the recovery tests are presented as Supplemental Data, Figure S5. A range of MMOAs can be applied in the DEBkiss model to explain the effects of stressors on the life history of an organism [23]. The time course for growth and reproduction in the different treatments indicates which MMOA(s) most likely underlies the observed toxic effects. In the present study, the data were described almost equally well when we assumed that acetone decreases assimilation (either by decreasing assimilation efficiency or feeding rate, resulting in a lower value of f), or when acetone increases the costs for both somatic and maturity maintenance by the same factor (higher value of J_M^V). This is a quite common situation, as both affected metabolic processes produce similar patterns of effects post-hatching [24]. In the present study we selected effects on assimilation efficiency because this MMOA was consistent with the observed effects on embryonic development (as explained in the section *Predictions for embryonic development*).

The model described the toxic effects over time well for all endpoints, even though only a small number of animals were used for calibration. The fits to data on survival (Figure 1A and B) and dry weight of soft body (Figure 1E and F) are less satisfactory, although it must be considered that there is only one data point per snail over the whole test duration (for survival, the time of death is the relevant endpoint).

We observed a hormetic response in the exposure tests. In the tests with juveniles, an increase in growth and reproduction compared with controls was observed, but not in the same treatment groups (0.1 mL L⁻¹ and 0.9 mL L⁻¹, respectively). Adult reproduction was also stimulated at the lowest concentrations. This indicates that low concentrations of acetone enhance snail performance, although this could well be an experimental artifact [25].

The data show that juvenile snails shrank at 18 mL L⁻¹ and adults shrank at 14 mL L⁻¹ to 18 mL L⁻¹ of acetone (Figure 1). Although the model predicted adult shrinking well, juvenile shrinking is generally underestimated. The reason why the

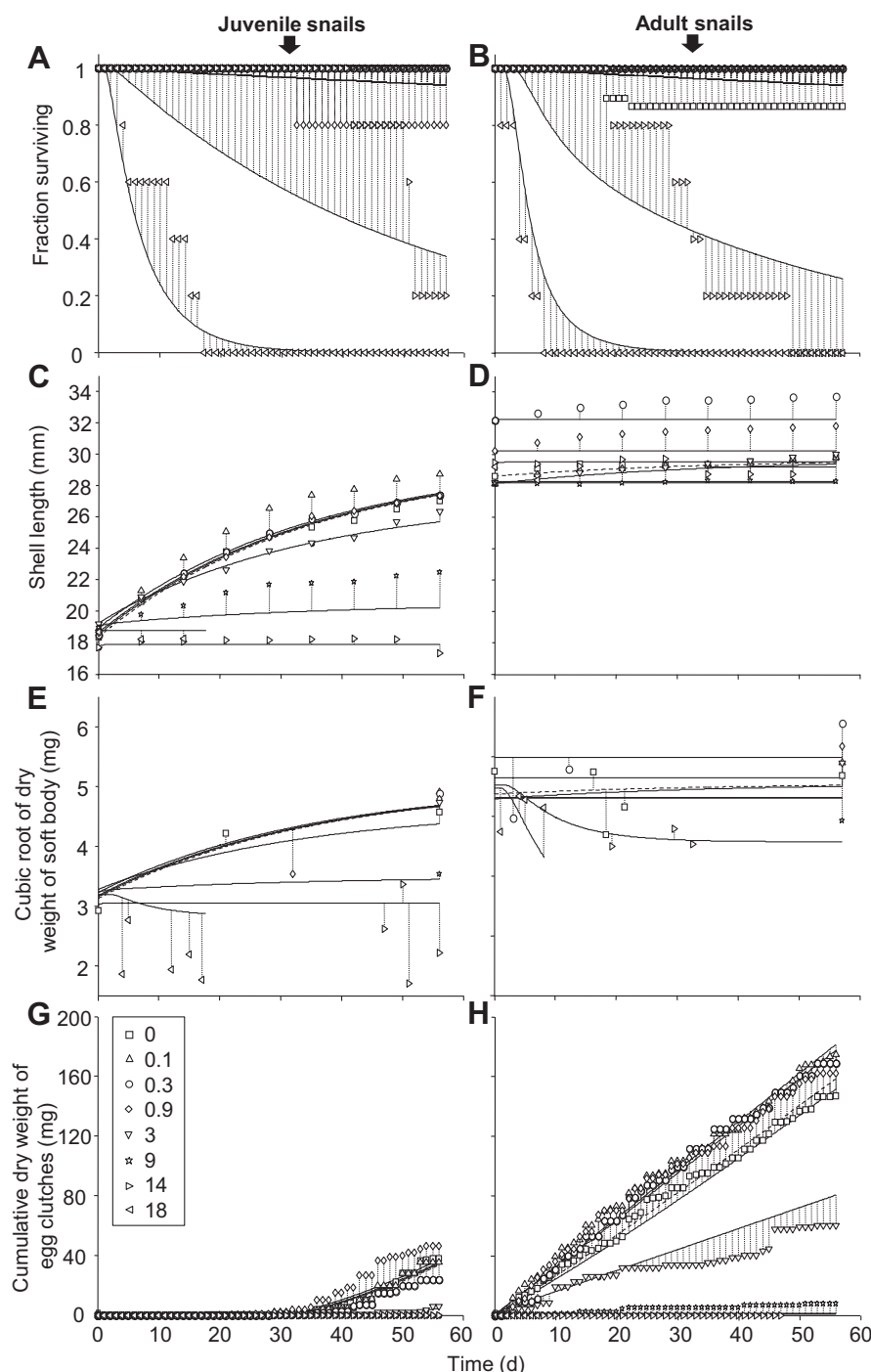


Figure 1. Model fits to effects of acetone in the exposure test on *Lymnaea stagnalis*. Fits to the following endpoints are shown: (A, B) survival, (C, D) shell length, (E, F) dry weight of soft body, (G, H) cumulative dry weight of egg clutches. Dotted and solid lines represent model curves for control and exposed snails, respectively. Symbols represent experimental data from different exposure concentrations of acetone (mL L^{-1}).

model does not capture adult and juvenile shrinking well with the same parameter set is unclear. This indicates that the model assumptions about starvation in *L. stagnalis* (see Supplemental Data) require further dedicated research.

In DEBkiss, toxic effects relate to the actual (scaled) internal concentration. Once snails are transferred to clean medium (in the recovery tests), the internal concentration will decrease over time, and thereby the effects. If an organism does not die, it is expected to fully recover after a certain period of time. The recovery patterns in the present study are well captured by the model, which shows that the assumption of full reversibility of effects on all endpoints holds for this compound (Supplemental

Data, Figure S5). To test assumptions about reversibility of effects, application of TKTD models is essential [26].

Comparing adequacy of various data sets for calibration

Next, we compared the values of toxicological parameters obtained by fitting different cross sections of the data set for exposed snails to investigate the value of different data types for model calibration. Most of the parameter values are shown in Figure 2, while a complete set of parameter values is presented in Supplemental Data, Figure S6. Interestingly, the parameter values are similar (as shown by overlapping confidence intervals) regardless of which data set was used for model

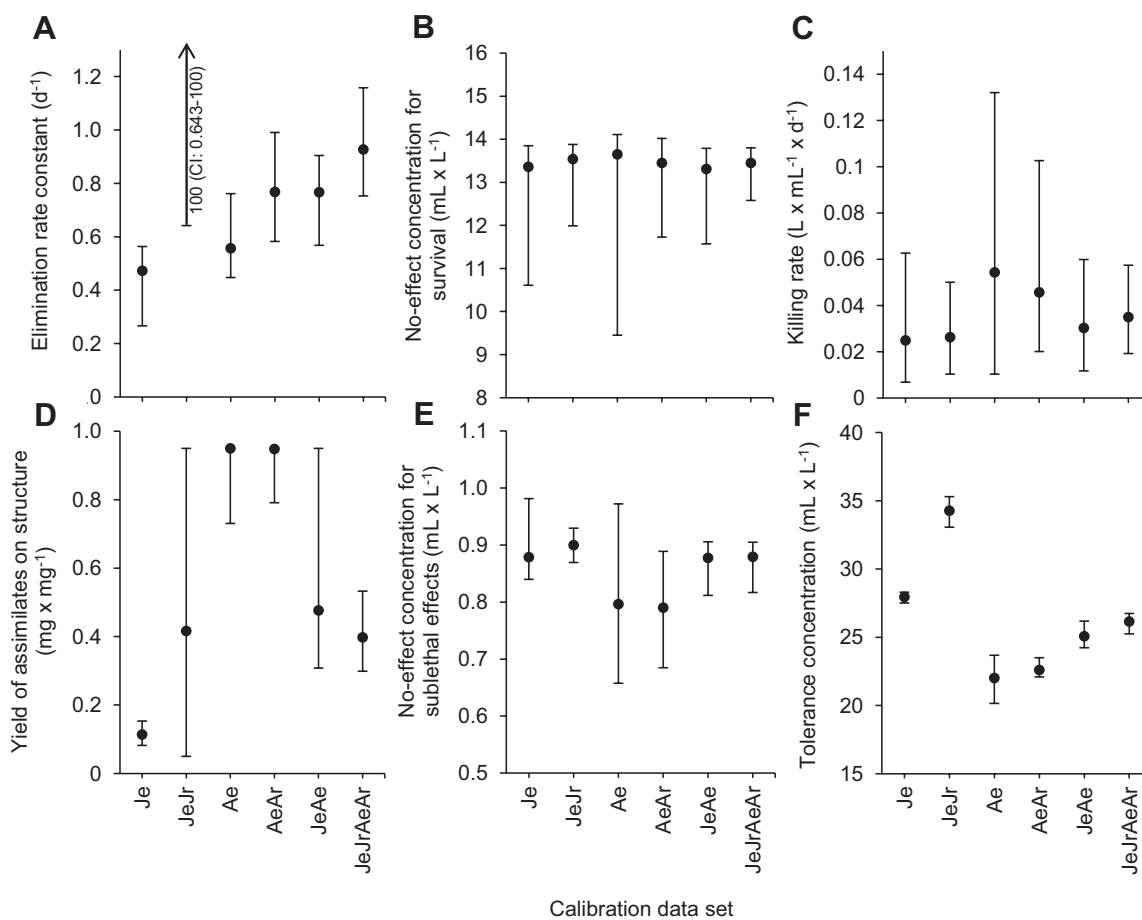


Figure 2. (A–F) Toxicological model parameter values estimated from various calibration data sets. Error bars represent 95% confidence intervals. The arrow in (A) indicates a very high parameter value (instantaneous steady state). Test abbreviations: Je = juvenile exposure; Ae = adult exposure; Jr = juvenile recovery; Ar = adult recovery.

calibration (Figure 2). The chosen calibration data set affects precision more than accuracy. Indeed, the confidence intervals of the toxicological parameters were narrowest when the model was calibrated on all data (the JeJrAeAr data set), which suggests that the data collected from all experiments provide a consistent picture of the toxicity of acetone. Overall, this analysis shows the robustness of the parameter estimation, as very different experimental sets lead to similar estimates for most parameters.

The best estimates for the NECs for survival and sublethal effects (Table 1) differ by a factor of 15.3, which ties in nicely with the 10-fold difference between commonly used trigger values of the toxicity exposure ratio in aquatic risk assessment (i.e., 10 for sublethal vs 100 for lethal effects [27]). The best estimate for the no-effect concentration values differs only slightly (overlapping confidence intervals) between the different calibration sets, which indicates that the model can robustly identify NEC values from very different data sets.

For the yield of assimilates on structure, there is a discrepancy between estimations obtained from the different data sets: data sets with adults only result in high yields, whereas juveniles favor low values (Figure 2). This yield coefficient is involved in the shrinking process, so this discrepancy is in line with the observation that juvenile shrinking is poorly captured by the model.

Subsequently, we compared how well the model parameters obtained from one data set would predict the observations in another (Figure 3). The general pattern is that a model calibrated

on one age class, regardless of the exposure type, predicts best the other data of that same age class. The model calibrated on the most complete data set within the age class (i.e., JeJr, AeAr, and JeJrAeAr) best predicted the less complete sets within the same class (i.e., Je and Jr, Ae and Ar, JeAe and JrAr). However, the precision with which data were predicted varies across the calibration sets. This analysis highlights the sensitivity of the model predictions toward differences in the values of toxicological parameters; small deviations in these parameter values as a result of a different calibration data set (Figure 2) may lead to significant differences in goodness of fit for predicted data sets (Figure 3). Given the difficulties in estimating y_{AV} , as discussed above, it is likely that the inadequacy of the starvation rules is the main culprit here.

Predictions for embryonic development

The metabolic parameters from Table 1 can be used to predict embryonic development in uncontaminated conditions. However, these parameters would lead to very short hatching times, as was also observed earlier [10]. Apparently, snail embryos do not grow much during the first few days, and then grow at a lower rate than expected from the growth curves of juveniles and adults. The first observation can be captured by a 2.5-d lag phase in development [10], but for the slow rate of development, the only explanation within DEBkiss is that the specific rate of utilization of the egg buffer is lower than the specific assimilation rate of juveniles and adults. Here, a factor of 0.4 explains the observed hatching time in the control (in Jager et al. [10], a factor

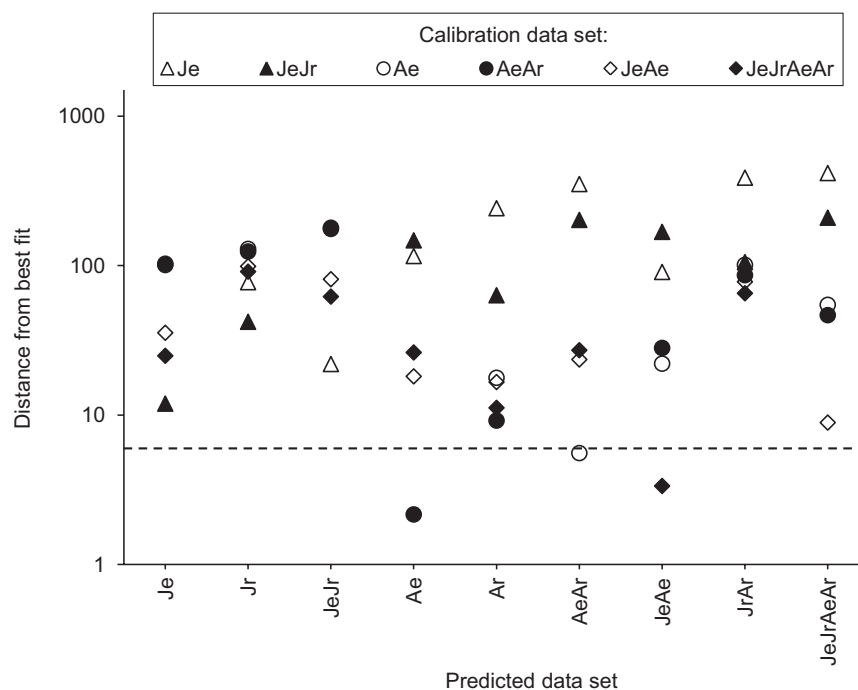


Figure 3. Data predicted from different calibration sets. The distance from the best fit represents the absolute difference in log-likelihood values between the prediction using a specific calibration set and the best possible fit (calibrating on the predicted data set itself). Lower values imply better predictions. Broken line represents the critical value of 6.3, above which predictions are significantly different from the best possible fit in a likelihood-ratio test at the 95% level. Test abbreviations: Je = juvenile exposure; Ae = adult exposure; Jr = juvenile recovery; Ar = adult recovery.

of 0.5 was used). This slower assimilation rate for embryos might be related to the apparent food limitation in early juveniles of the pond snail [24]; juveniles up to a certain shell size grow more slowly than later in life, which relates to food quality and/or feeding rates. The predicted absolute shell size at hatching (not shown) is much larger than the observed values. This likely is a result of using the same shape coefficient for hatchlings as for adults (shells of hatchlings are less pointy). For this reason, in Figure 4A we present shell size relative to that of control snails.

The predictions for the embryonic endpoints as a function of the acetone concentration are provided in Figure 4A through C for the 3 MMOAs that explain juvenile and adult data well: a decrease in feeding rate (i.e., a decrease in the utilization rate of assimilates by the embryos from the egg buffer), a decrease in

assimilation efficiency, and a combined increase in somatic and maturity maintenance costs. The first 2 MMOAs are both reflected in a change of the value of the parameter f for the embryo, just as for juveniles and adults. However, in contrast to juveniles and adults, food is not supplied ad libitum for the embryo, which leads to differences in development between the 2 MMOAs. Decreasing the utilization rate of the limited egg buffer means that the nonassimilated food is available for later use, whereas a decrease in the assimilation efficiency implies an increase in the overhead costs and thus that a larger part of the egg buffer is dissipated. Interestingly, the hatching time and hatchling size allowed us to distinguish between various possible MMOAs of acetone. A decrease in the assimilation efficiency clearly provided the best explanation for the observed patterns.

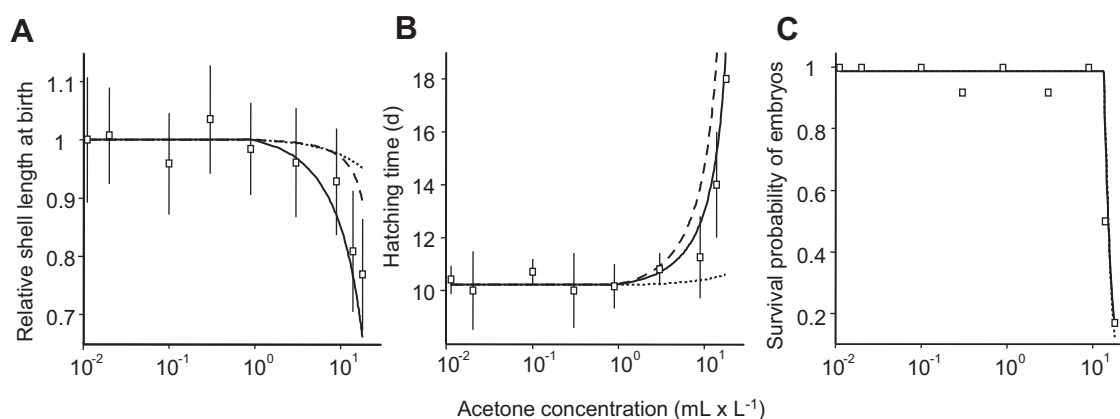


Figure 4. Predictions of embryo responses to acetone: (A) shell length at birth relative to control, (B) hatching time, (C) survival probability at day 14 of the test. Squares represent data points. Error bars represent standard deviations of the data. Curves represent predictions with different metabolic mode of action: solid line, decrease of assimilation efficiency; broken line, decrease of feeding rate (i.e., utilization of the egg buffer); dotted line, increase of somatic and maturity maintenance costs.

Selecting the most probable MMOA for a toxicant is important from a scientific point of view and also for practical purposes (especially in the extrapolation to other environmental conditions), but is a recurring problem in the application of DEB models in ecotoxicology [28,29]. In many cases, standard toxicity tests do not provide enough information to differentiate between various MMOAs (especially assimilation and maintenance). Focusing on goodness-of-fit can be dangerous, as the best fitting MMOA may well depend on rather arbitrary assumptions in the statistical approach (e.g., transformation of the data). Here, we demonstrate how effects on egg development can aid the selection of the MMOA.

Prediction of the toxic effect of acetone on embryonic development from the parameter values in Table 1 suggests that embryos are not intrinsically more or less sensitive than juveniles and adults to this compound (where intrinsic sensitivity is defined as the relationship between the internal concentration and a metabolic process [26]). In addition, the predictions show that the predicted survival probability for the embryo corresponds well with the observed hatching success. Even though we did not check whether nonhatched embryos were alive or not, the acetone treatments in which the NEC for survival is exceeded result in a drop of the hatching success with the same factor as the predicted survival probability. Finally, we could predict both embryonic development and juvenile and adult life-cycle traits by the same MMOA—a decrease in assimilation efficiency. This supports the DEBkiss interpretation of embryonic development (where embryos assimilate in the egg) in contrast to the standard DEB model assumption (where embryos do not assimilate but mobilize their reserve), at least for the pond snail. It should be noted that toxicity tests on eggs extracted from their jelly-mass [20] are inappropriate to realistically assess the effects on embryos of *L. stagnalis* in the field. However, to design a simple TKTD model, it is important to remove the complicating factors that the egg mass contributes to toxicokinetics but also to the development of the embryos [30].

Consequences for experimental design

Overall, DEBkiss described the effects of acetone well, with rather similar parameter values irrespective of the calibration data provided. In the present study, we estimated the metabolic parameters using the control data for juveniles and adults together. It must be stressed that data from juveniles only, and especially adults only, is not sufficient to parameterize TKTD models such as DEBkiss because neither data set by itself contains enough information to first calibrate the model for control (uncontaminated) conditions. To estimate the metabolic parameters, data on the growth curve, the start of reproduction, and the reproduction rate are required, which implies testing with juveniles and adults (or following juveniles well into adulthood). Another limitation of using data on adult snails only is that it becomes more difficult to identify a single, most probable MMOA for the toxicant; effects on the growth curve are essential to distinguish between the various options. Furthermore, as adult snails are close to their maximum size at the start of the experiment, a toxic effect on assimilation or maintenance will rapidly induce starvation, making it difficult to decide whether a poor fit is the result of an incorrect MMOA or unrealistic starvation rules.

Adding information on recovery made surprisingly little improvement on the precision of the parameter estimates. Only the confidence intervals of the NEC and the killing rate were slightly narrower when data on recovery were included. In

contrast, it has been shown that in some cases including exposure–recovery treatment can substantially improve the precision of some parameter estimates [31]. From a risk assessment perspective, it might be important to test the reversibility of effects, particularly for plant protection products, whose application often results in highly time-variable exposure. In that case, it is important to assess whether or not individuals fully recover between pulses or whether damage accumulates [32].

We further show that the additional determination of dry weight at the end of the test is extremely helpful to establish whether snails shrink in the treatments, which is clearly not possible from shell size only. As the dry weight is a destructive measurement, it cannot be used to monitor changes in weight of exposed animals along the test. It would be interesting to investigate whether measuring wet body weight (including the shell), at least several times during the test, could provide a useful source of information. Obtaining weight data also helps to decrease the options for the MMOA in some cases (only effects on assimilation and somatic maintenance will induce shrinking).

Reproductive performance in toxicity tests is usually measured as number of eggs. However, DEBkiss specifies a reproductive mass flux as the primary output of the model. Thus, it is more accurate to calibrate the model directly on data for dry mass of clutches than to count the number of eggs and use an average dry mass per egg. Weighing egg clutches instead of counting eggs also requires less effort. However, counting the eggs in at least some clutches is important to identify possible effects on the investment of assimilates per egg (which would be essential to assess the population consequences of the stressor).

The standard OECD protocol under development for *L. stagnalis* includes 30 snails per concentration, which is recommended when a descriptive analysis of the data is foreseen, given the large variability of the snails. Calibrating the DEBkiss model makes no strict demands on the number of individuals; increasing the number of individuals increases the reliability of the analysis and leads to narrower confidence intervals on the parameters. The number of animals thus has to be chosen based on the desired precision and the variability in the test population. In the present study we demonstrate that even with 5 animals per treatment we were able to estimate the NECs with good precision (Figure 2). In our experimental setup, we followed individual snails rather than keeping them in groups. Even though we did not make use of this property in the present study, it allows for analyses accounting for interindividual variability [33], which could further increase the efficiency of model calibration.

The TKTD models also make no strict demands on the number of test concentrations, although a range of effects (from none to large) is recommended to allow identification of the toxicological parameters and a unique MMOA. To calibrate such models, it is, however, absolutely essential to have observations over time for multiple endpoints (body size, reproduction, survival). Statistical power can most easily be increased by increasing the number of observations over time.

In summary, the most appropriate test design to calibrate the DEBkiss model would be to follow all endpoints over time from the same cohort, starting from the juvenile stage and continuing well into adulthood. For body size, shell length and dry body weight (at the end of the test) are suitable endpoints, although additional wet-weight measurements might be useful. For reproduction, the dry weight of clutches is most appropriate, with additional counting of the number of eggs in some clutches. To select a most appropriate MMOA, data on some additional

endpoints might be helpful, such as the feeding rate [34] or data on embryonic development, as demonstrated in the present study. Testing recovery does not specifically help the estimation of model parameters (at least in this case study), but is very helpful to demonstrate the reversibility of effects.

CONCLUSIONS

The DEBKiss model can be used to explain the toxic effects of acetone in *L. stagnalis* on survival, growth, and reproduction over time, in one integrated analysis. The model also provided a consistent explanation for effects on embryonic development. In fact, the embryo data enabled us to decide on the most appropriate MMOA, which opens up possibilities of including embryonic effects (which are often ignored in ecotoxicology) as an integrated element in test design. We also showed that the intrinsic sensitivity of snails to acetone stays the same across all developmental stages, which may well be a feature of narcotic chemicals in general.

Clearly, calibration of TKTD models such as DEBKiss places different demands on test design compared with classical concentration–response approaches. The draft OECD protocol for the pond snail is thus not optimally suited to calibrate TKTD models, and is also not intended for that purpose. In the present study we discuss a range of practical considerations for test design relating to the calibration of TKTD models. However, one should realize that different toxicants may require different designs. For substances other than narcotics, differences between life stages may be more important, and recovery may be less predictable. For TKTD models, a flexible test design does not provide a fundamental obstacle: as our analysis shows, different calibration data will tend to affect precision (the confidence intervals) rather than accuracy of parameter estimates. More specific recommendations on test design will depend on the purpose of the analysis: for the estimation of a single parameter (such as the NEC), optimal test design is different from designing for the link to population models [35].

SUPPLEMENTAL DATA

Sections 1–2. (402 KB PDF).

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REFERENCES

- Giusti A, Barsi A, Dugué M, Collinet M, Thomé J-P, Joaquim-Justo C, Roig B, Lagadic L, Ducrot V. 2013. Reproductive impacts of tributyltin (TBT) and triphenyltin (TPT) in the hermaphroditic freshwater gastropod *Lymnaea stagnalis*. *Environ Toxicol Chem* 32: 1552–1560.
- International Organization for Standardization. 2006. Water quality. Guidance on statistical interpretation of ecotoxicity data. ISO 20281. Berlin, Germany.
- Organization for Economic Cooperation and Development. 2006. Current approaches in the statistical analysis of ecotoxicity data: A guidance to application. Series on testing and assessment 54. Paris, France.
- Ashauer R, Escher BI. 2010. Advantages of toxicokinetic and toxicodynamic modelling in aquatic ecotoxicology and risk assessment. *J Environ Monit* 12:2056–2061.
- Jager T, Heugens EW, Kooijman SALM. 2006. Making sense of ecotoxicological test results: towards application of process-based models. *Ecotoxicology* 15:305–314.
- Hunka AD, Meli M, Thit A, Palmqvist A, Thorbek P, Forbes VE. 2013. Stakeholders' perspective on ecological modeling in environmental risk assessment of pesticides: Challenges and opportunities. *Risk Analysis* 33:68–79.
- European Commission. 2012. Preliminary report on addressing the new challenges for risk assessment. Brussels, Belgium.
- Organization for Economic Co-operation and Development. 2010. Detailed review paper on molluscs life-cycle toxicity testing. Series on Testing and Assessment, No. 121. ENV/JM/MONO (2010) 9. Paris, France.
- Kooijman SALM. 2001. Quantitative aspects of metabolic organization: A discussion of concepts. *Philos T Roy Soc B* 356:331–349.
- Jager T, Martin BT, Zimmer EI. 2013. DEBKiss or the quest for the simplest generic model of animal life history. *J Theor Biol* 328:9–18.
- Jager T, Zimmer EI. 2012. Simplified dynamic energy budget model for analysing ecotoxicity data. *Ecol Model* 225:74–81.
- Kooijman SALM, Bedaux JJM. 1996. Analysis of toxicity tests on fish growth. *Water Res* 30:1633–1644.
- Kooijman SALM, Bedaux JJM. 1996. Analysis of toxicity tests on *Daphnia* survival and reproduction. *Water Res* 30:1711–1723.
- Sousa T, Domingos T, Poggiale J-C, Kooijman SALM. 2010. Dynamic energy budget theory restores coherence in biology. *Philos T Roy Soc B* 365:3413–3428.
- Alda Álvarez O, Jager T, Redondo EM, Kammenga JE. 2006. Physiological modes of action of toxic chemicals in the nematode *Acrobeloides nanus*. *Environ Toxicol Chem* 25:3230–3237.
- Jager T, Albert C, Preuss TG, Ashauer R. 2011. General unified threshold model of survival—A toxicokinetic-toxicodynamic framework for ecotoxicology. *Environ Sci Technol* 45:2529–2540.
- Jumel A, Coutellec M-A, Cravedi J-P, Lagadic L. 2002. Nonylphenol polyethoxylate adjuvant mitigates the reproductive toxicity of fomesafen on the freshwater snail *Lymnaea stagnalis* in outdoor experimental ponds. *Environ Toxicol Chem* 21:1876–1888.
- Bluzat R, Jonot O, Lespinasse G, Seugé J. 1979. Chronic toxicity of acetone in the fresh water snail *Lymnaea stagnalis*. *Toxicology* 14:179–190.
- Hoffer JNA, Schwegler D, Ellers J, Koene JM. 2012. Mating rate influences female reproductive investment in a simultaneous hermaphrodite, *Lymnaea stagnalis*. *Anim Behav* 84:523–529.
- Bandow C, Weltje L. 2012. Development of an embryo toxicity test with the pond snail *Lymnaea stagnalis* using the model substance tributyltin and common solvents. *Sci Total Environ* 435–436:90–95.
- van Duivenboden YA. 1983. Transfer of semen accelerates the onset of egg-laying in female copulants of the hermaphrodite freshwater snail, *Lymnaea stagnalis*. *Int J Inver Rep* 6:249–257.
- Escobar JS, Auld JR, Correa AC, Alonso JM, Bony YK, Coutellec M-A, Koene JM, Pointier J-P, Jarne P, David P. 2011. Patterns of mating-system evolution in hermaphroditic animals: Correlations among selfing rate, inbreeding depression, and the timing of reproduction. *Evolution* 65:1233–1253.
- Jager T. 2013. *Making Sense of Chemical Stress: Applications of Dynamic Energy Budget Theory in Ecotoxicology and Stress Ecology*, version 14. [cited 2013 October 14]. Available from: <http://www.debttox.info/book.php>.
- Zimmer E, Jager T, Ducrot V, Lagadic L, Kooijman SALM. 2012. Juvenile food limitation in standardized tests: A warning to ecotoxicologists. *Ecotoxicology* 21:2195–2204.
- Jager T, Barsi A, Ducrot V. 2013. Hormesis on life-history traits: Is there such thing as a free lunch? *Ecotoxicology* 22:1–8.
- Pieters B, Jager T, Kraak MS, Admiraal W. 2006. Modeling responses of *Daphnia magna* to pesticide pulse exposure under varying food conditions: Intrinsic versus apparent sensitivity. *Ecotoxicology* 15:601–608.
- European Commission. 2002. Guidance document on aquatic ecotoxicology in the context of the Directive 91/414/EEC. SANCO/3268/2001. Brussels, Belgium.
- Martin B, Jager T, Nisbet R, Preuss T, Hammers-Wirtz M, Grimm V. 2013. Extrapolating ecotoxicological effects from individuals to populations: A generic approach based on dynamic energy budget theory and individual-based modeling. *Ecotoxicology* 22:574–583.

29. Muller E, Osenberg C, Schmitt R, Holbrook S, Nisbet R. 2010. Sublethal toxicant effects with dynamic energy budget theory: Application to mussel outplants. *Ecotoxicology* 19:38–47.
30. Marois R, Croll RP. 1991. Hatching asynchrony within the egg mass of the pond snail, *Lymnaea stagnalis*. *Invertebr Reprod Dev* 19:139–146.
31. Albert C, Ashauer R, Künsch HR, Reichert P. 2012. Bayesian experimental design for a toxicokinetic–toxicodynamic model. *J Stat Plan Inference* 142:263–275.
32. Ashauer R, Boxall A, Brown C. 2006. Predicting effects on aquatic organisms from fluctuating or pulsed exposure to pesticides. *Environ Toxicol Chem* 25:1899–1912.
33. Jager T. 2013. All individuals are not created equal: Accounting for interindividual variation in fitting life-history responses to toxicants. *Environ Sci Technol* 47:1664–1669.
34. Zimmer EI. 2013. The pond snail under stress: Interactive effects of food limitation, toxicants and copulation explained by Dynamic Energy Budget theory. PhD thesis. VU University, Amsterdam, The Netherlands, pp 33–40.
35. Jager T, Barsi A, Hamda NT, Martin BT, Zimmer EI, Ducrot V. 2014. Dynamic energy budgets in population ecotoxicology: Applications and outlook. *Ecol Model* 280:140–147.